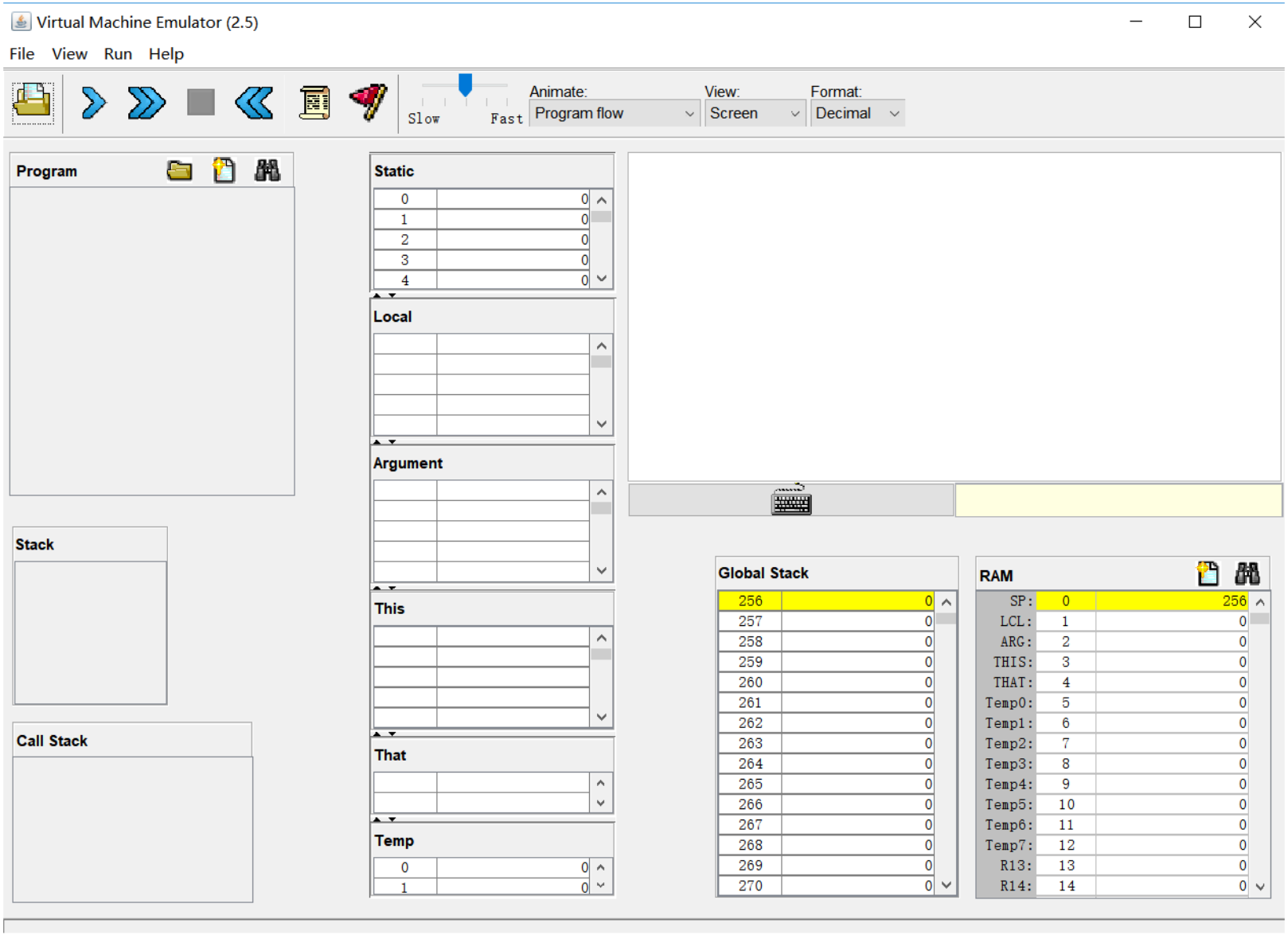


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UK | CHINA | MALAYSIA

Virtual Machine (Part 1)

Dr. Wooi Ping Cheah

Introduction to VMEmulator



VMEmulator Memory Layout

- Pointers: RAM[0 … 4]

  Base address for various memory segments: Global Stack (SP), Local (LCL), Argument (ARG), THIS, THAT
- Temp0 –Temp7: RAM[5 … 12]
- General: RAM[13 … 15]
- Static: RAM[16 ... 255]
- Stack: RAM[256 ... 2047]
  Global Stack, Local, Argument
- Heap: RAM[2048 onward]

  Dynamic memory locations pointed with pointers THIS (pointer 0) and THAT (pointer 1)

How to test VM codes?

- Two ways to test VM codes:

  - Manually set the base memory address first before you test the VM codes, e.g.
    - set sp 256, // stack pointer
    - set local 300, // base address of the local segment
    - set argument 400, // base address of the argument segment
    - set this 3000, // base address of the this segment
    - set that 3010, // base address of the that segment

  - Use test scripts to test the VM codes.

    e.g. **BasicTest.tst**

Task 1: Tracking Stack Status

- Given the following VM codes, manually track the stack status after each stack operation. What is the final status of the stack?
  - SimpleAdd.vm,
  - StackTest.vm,
  - BasicTest.vm,
  - StaticTest.vm,
  - PointerTest.vm.
- Verify your answers by running the VM codes on the VMEmulator.

Examples of Bitwise Operations

- Examples of bitwise operations found in **StackTest.vm**

Notation:

1's – 1's Complement

2's – 2's Complement

Boolean Value Representation:

True – -1 (Binary: 1111111111111111)

False – 0 (Binary: 0000000000000000)

Examples of Bitwise Operations

- Examples of bitwise operations found in **StackTest.vm**

57 and 28 = 24

57 0000000000111001

28 0000000000011100

and 0000000000011000 = 24

24 or 82 = 90

24 0000000000011000

82 0000000001010010

or 0000000001011010 = 90

Examples of Bitwise Operations

- Examples of bitwise operations found in **StackTest.vm**

57 and 0 = 0

57 0000000000111001

0 0000000000000000

and 0000000000000000 = 0

57 or -1 = -1

57 0000000000111001

-1 1111111111111111

or 1111111111111111 = -1

Examples of Bitwise Operations

- Examples of bitwise operations found in **StackTest.vm**

not 90 = -91

90 0000000001011010

not 1111111110100101 = -91

-(-91)

-91 1111111110100101

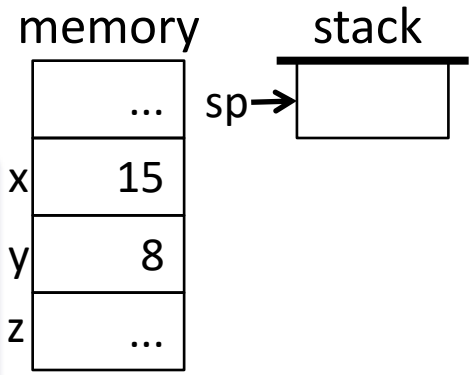
1's 0000000001011010

2's 0000000001011011 = 91

Task 2: Show Stack Operations

VM code

.....
.....
.....
.....
.....
.....
.....
.....



Show the stack operations for the following expression:

z = x > 8 and y < 8

What is the final memory status?

Task 3: Write VM Loop Function

- Write a basic loop function in VM code.
  - Computes the sum 1 + 2 + … + Argument[0], and pushes the result onto the stack.
- Test the VM code on VMEmulator.
- You may refer to the sample code called ***mult*** for multiplying two numbers.

Acknowlegement

- This set of lecture notes are based on the lecture notes provided by Noam Nisam / Shimon Schocken.
- You may find more information on: www.nand2tetris.org.